Power and water control for Nigerian buildings.

A.D.A. is a small controller and dashboard that knows whether a building is on NEPA, generator or inverter, logs every generator hour, runs the borehole pump on tank level, sheds heavy loads on inverter, and protects equipment from voltage spikes. Every automatic action is recorded with its reason.

THE PROBLEM IT SOLVES

Diesel	No reliable record of generator hours, so reconciliation is guesswork.
Water	Pumps left running overflow tanks; forgotten pumps leave tanks dry.
Batteries	Heavy loads on inverter drain batteries overnight and shorten their life.
Equipment	Voltage spikes damage TVs, fridges, CCTV and access control.

WHAT IT DOES

Live source	NEPA, gen, inverter or none, with voltage, for every building.
Gen & diesel log	Run-hours per generator per day, with estimated diesel.
Borehole	Starts on low tank when power is available, stops at 95%.
Protection	Sheds heavy loads on inverter; cuts sensitive loads on spikes.
Forecasts	Battery runtime, usual outage times, days of diesel and units left.
Ask A.D.A.	Explains what happened and why, in English or Pidgin.

Works offline

Rules, log and forecasts run on site. Internet adds remote access and the full assistant.

Fails safe

If the controller loses the hub for 10 s, relays drop to safe states. Stale data is never shown as live.

Fits what you have

Works alongside the existing generator, inverter and changeover. Mains wiring by a qualified electrician.

STATUS & PROPOSAL

Working prototype: ESP32 controller reporting live to the hub, automation rules running, dashboard built. **Proposal:** a pilot on one Max-Migold building for about six weeks, to record its power and water, automate the routine switching, and review the results together.

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